

Cryptology Assignment - 3

① Determine the multiplicative inverse module 26 for each of the following integers.

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

$$3^{-1} \bmod 26$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1) \quad \text{--- (1)}$$

$$2 = 26 + 3(-8) \quad \text{--- (2)}$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(9) + 26(-1) = 1$$

$$\underline{\underline{3^{-1} = 9}}$$

$$5^{-1} \bmod 26$$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$\underline{\underline{5^{-1} = 21}}$$

$$7^{-1}$$

$$26 = 7(3) + 5$$

$$5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$5 + 2[7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$\underline{\underline{7^{-1} = 15}}$$

$$9^{-1}$$

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$\underline{\underline{9^{-1} = 3}}$$

11⁻¹

$$26 = 11(2) + 4 \quad 4 = 26 + 11(-2)$$

$$11 = 4(2) + 3 \quad 3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$11^{-1} = 19$$

~~13~~ 15⁻¹

$$26 = 15(1) + 11 \quad 11 = 26 + 15(-1)$$

$$15 = 11(1) + 4 \quad 4 = 15 + 11(-1)$$

$$11 = 4(2) + 3 \quad 3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + (11 + 4(-2))(-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(3) + 26(-2) + (26 + 15(-1))(-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$15^{-1} = 7$$

17^{-1}

$$26 = 17(1) + 9 \quad 9 = 26 + 17(-1)$$

$$17 = 9(1) + 8 \quad 8 = 17 + 9(-1)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + (17 + 9(-1))(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$\underline{\underline{17^{-1} = 23}}$$

 19^{-1}

$$26 = 19(1) + 7 \quad 7 = 26 + 19(-1)$$

$$19 = 7(2) + 5 \quad 5 = 19 + 7(-2)$$

$$7 = 5(1) + 2 \quad 2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$19 + 7(-2) + (7 + 5(-1))(-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + 7(-4) + 5(2) = 1$$

$$19 + (26 + 19(-1))(-4) + (19 + 7(-2))(2) = 1$$

$$19 + 26(-4) + 19(+4) + 19(2) + 7(-4) = 1$$

$$19 + 26(-4) + 26(-4) + 19(4) = 1$$

$$19(1) + 26(-8) = 1$$

$$\underline{\underline{19^{-1} = -8}}$$

21^{-1}

$$\cancel{26 = 21(1) + 5} \quad \cancel{6 = 26 + 21(-1)}$$

$$\cancel{21 = 5(3) + 2}$$

$$26 = 21(1) + 5$$

$$5 = 26 + 21(-1)$$

$$21 = 5(4) + 1$$

$$1 = 21 + 5(-4)$$

$$21 + 5(-4) = 1$$

$$21 + (26 + 21(-1))(-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$26(-4) + 21(5) = 1$$

$$21^{-1} = 5$$

 23^{-1}

$$26 = 23(1) + 3$$

$$3 = 26 + 23(-1)$$

$$23 = 3(7) + 2$$

$$2 = 23 + 3(-7)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + (23 + 3(-7))(-1) = 1$$

$$26 + 23(-1) + 23(-1) + (26 + 23(-1))(-7) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$23^{-1} = 17$$

$$25^{-1}$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$\underline{25^{-1} = 25}$$

- ② Choose any string of your choice to encrypt using the ✓ affine cipher with key value: multiplier 5 & additive 8.

PLAINTEXT: AANSH

0 0 13 18 7

Formula: $C = (P \cdot \text{Multi} + \text{Add}) \bmod 26$

$$C = (P \times 5 + 8) \bmod 26$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 \quad I$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 \quad I$$

$$N = (13 \times 5 + 8) \bmod 26 = (65 + 8) \bmod 26 = (73) \bmod 26 = 21 = V$$

$$S = (18 \times 5 + 8) \bmod 26 = (90 + 8) \bmod 26 = 98 \bmod 26 = 18 = S$$

$$H = (7 \times 5 + 8) \bmod 26 = (35 + 8) \bmod 26 = 43 \bmod 26 = 17 = R$$

CIPHERTEXT = IIVSR

Encrypt 'HI' using affine cipher with $A=11, b=5$

$$C = (P \times a + b) \bmod 26$$

$$H = 7$$

$$C = (7 \times 11 + 5) \bmod 26$$

$$I = 8$$

$$C = H = (7 \times 11 + 5) \bmod 26 = (77 + 5) \bmod 26 = 82 \bmod 26 = 4 = E$$

$$\underline{I} = (8 \times 11 + 5) \bmod 26 = (88 + 5) \bmod 26 = 93 \bmod 26 = 15 = P$$

$$\text{Ciphertext} = EP$$

The Ciphertext UCR was encrypted using the affine function $(9x+2) \bmod 26$ find the plaintext
 $U=20, C=2, R=17$

$$C = (9P + 2) \bmod 26$$

$$P = a^{-1} (C - b) \bmod 26$$

$$a = 9$$

$$b = 2$$

$$a^{-1}$$

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

$$U = 39(20 - 2) \bmod 26 = 39(18) \bmod 26 = 702 \bmod 26 = 54 \quad 2=C$$

$$C = 39(2 - 2) \bmod 26 = 39(0) \bmod 26 = 0 = A$$

$$R = 39(17 - 2) \bmod 26 = 39(15) \bmod 26 = 585 \bmod 26 = 45 \quad 19=T$$

$$\text{Plain text} = CAT$$

5) Decrypt the ciphertext using affine cipher
 K D Y X J U W D E X Z Y using the key 3(mod 26).

$$a = 3 \quad b = 4$$

$$P = a^{-1}(C - b) \bmod 26$$

K D Y X J U W D E X Z Y
 10 3 24 23 9 20 22 3 4 23 25 24

$$a^{-1} = 3^{-1}$$

$$26 = 3(8) + 2 \quad 2 = 26 + 3(-8)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + (26 + 3(-8))(-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$\underline{\underline{3^{-1} = 9 = a^{-1}}}$$

$$K = 9(10 - 4) \bmod 26 = 9(6) \bmod 26 = 2 = C$$

$$D = 9(3 - 4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$Y = 9(24 - 4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

$$X = 9(23 - 4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$J = 9(9 - 4) \bmod 26 = 9(5) \bmod 26 = 19 = T$$

$$U = 9(20 - 4) \bmod 26 = 9(16) \bmod 26 = 14 = O$$

$$W = 9(22 - 4) \bmod 26 = 9(18) \bmod 26 = 6 = G$$

$$D = 9(3 - 4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$E = 9(4 - 4) \bmod 26 = 9(0) \bmod 26 = 0 = A$$

$$X = 9(23 - 4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$Z = 9(25 - 4) \bmod 26 = 9(21) \bmod 26 = 7 = H$$

$$Y = 9(24 - 4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

Plaintext : CRYPTOGRAPHY :

- ⑥ Use $K_1 = 13$ (Multi), $K_2 = 4$ (Even though it is not a valid key - try to encrypt). Encrypt the string 'input' and 'alter' above key & find what is issue.

$$C = (P \times \text{Multi} + \text{Add}) \bmod 26$$

$$C = (P \times a + b) \bmod 26$$

$$a = 13 \quad b = 4$$

13 is not coprime of 26.

So we can't find inverse of 13 mod 26.

Therefore, it is not a valid key for affine cipher.

For 'input'

$$I = (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 = R$$

$$P = (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$U = (20 \times 13 + 4) \bmod 26 = 264 \bmod 26 = 4 = E$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

ERRER

For 'alter'

$$A = (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 = E$$

$$L = (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 = R$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

$$E = (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 = E$$

$$R = (17 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 = R$$

ERRER

Both the plaintext will produce the same ciphertext. The encryption is not one to one and decryption is not possible as 13 is not a valid key.

⑦ Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \pmod{26}$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \pmod{26}$$

$$\det(K) = 3 \times 7 - 2 \times 5 = 21 - 10 = 11$$

$$11^{-1}$$

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$~~26(-3)~~ 26(3) + 11(-7) = 1$$

$$~~11(-7)~~ 26 - 7 = 19$$

Adjacent of the matrix

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following matrix whose entries are considered modulo 26: $K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \text{ mod } 26$$

$$\det(K) = 11 \times 3 - 13 \times 2 = 33 - 26 = 7$$

7^{-1}

$$26 = 7(3) + 5$$

$$5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$26 + 7(-3) + 2(7 + 5(-1))(-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + (26 + 7(-3))(2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

Adjacent is

$$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9) Is the key valid?

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

For a key to be valid - its determinant should be coprime and shouldn't be equal to 0

$$\det(k) = 2(21) - 14(13)$$

$$= 42 - 182$$

$$= -140$$

$$-140 \bmod 26 = 16$$

$$\det k = 16 \bmod 26$$

16 is not coprime to 26

Therefore it will not have an inverse module 26

$\therefore$ The key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid.

(10) Encrypt Using 2×2 Hill Cipher for the plaintext
'SOLVED' Using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$3 \times 1 - 7 \times 8 = -23 \equiv 3 \text{ which is coprime}$$

$$S O L V E D \quad C = K \times P$$

$$P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \text{ mod } 26$$

$$166 \text{ mod } 26 = 10 = K$$

$$280 \text{ mod } 26 = 20 = U$$

$$\begin{matrix} L \\ V \end{matrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \text{ mod } 26$$

$$201 \text{ mod } 26 = 19 = T$$

$$308 \text{ mod } 26 = 22 = W$$

$$\begin{matrix} E \\ D \end{matrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \text{ mod } 26$$

$$36 \text{ mod } 26 = 10 = K$$

$$61 \text{ mod } 26 = 9 = J$$

SOLVED $\rightarrow$ KUTWKJ